

A COMPLETE BEGINNER'S COURSE

The Practical Data Analyst's Handbook

From Zero to Your First Data Job in
Excel, SQL, Statistics, Python and Power BI

Tatenda Makuva

"The Big Data Analyst"

First Edition · 2026

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Front Matter

Welcome to Your New Career

The Practical Data Analyst's Handbook — From Zero to Your First Data Job: A Complete Beginner's Course

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First Edition, 2026

Who This Book Is For

You have never written a line of code. You open a spreadsheet and feel slightly tired. The word "statistics" reminds you of a classroom you were happy to leave. And yet somewhere you have heard that data analysis is a good, honest, well-paid skill — one that exists in every industry, one you could actually learn — and you wonder whether someone like you could really do it.

Yes. This book was written precisely for you.

It assumes **zero background**. No mathematics beyond arithmetic. No programming. No prior tools. Every term is explained the moment it appears, in plain English, with a worked example before you are asked to try anything yourself. Every tool in this book has a free version, and Chapter 4 shows you how to install all of them in one afternoon.

The Promise

By the last page you will have:

1. **Five tools, genuinely learned** — Excel, SQL, statistics (with SPSS and Stata), Python, and Power BI — each introduced from absolute zero, each practiced on real (generated) business data.
2. **Six completed projects** — including three portfolio-ready analyses with walkthroughs — all built on the running case of a real-seeming small business.
3. **A portfolio** — a GitHub page an employer can click, showing your work, by Chapter 49.
4. **The job hunt, handled** — CV, LinkedIn, technical interviews, case interviews, take-homes, and your first 90 days once hired.
5. **The map of what comes next** — machine learning, big data, ethics, and the career decade after your first job.

How This Book Works

One running case. In Chapter 5 you meet **Tariro**, who runs a grocery store in Highfield, Harare — a notebook, a half-trusted spreadsheet, and a dream of a second branch. You become her analyst, and you stay hired for the whole book. Every formula, query, and chart you learn is immediately used on her real-seeming data (generated with documented patterns, so your answers can always be checked against the truth). When you interview for your first job, Tariro is your story.

One chapter recipe. Every chapter tells you what you will learn, teaches it with worked examples, closes with Key Takeaways and a Practice Lab you can actually do (data provided), and points you to further reading. Short chapters, steady stairs.

One toolkit, bridged. The "From Your Toolkit" callouts connect every new tool back to the ones you already know — the same table-join appears first in Excel, then in SQL, then in Python — so nothing you learn is ever an island.

One honesty habit. Averages lie, samples have error, correlation is not causation. You will meet these truths in Part I and be held to them in every part after. The most employable thing about you will not be a tool; it will be that your numbers can be trusted.

The Journey

Part	Question it answers	Tools
I — Beginning	What does a data analyst actually do, and how do I start?	The mindset; the workspace
II — Excel	How do I turn a messy sheet into a report?	Excel, PivotTables, charts
III — SQL	How do I ask a database real questions?	SQL, joins, windows
IV — Statistics	How do I know a number means anything?	Descriptives, tests, SPSS, Stata
V — Python	How do I make the computer do the repetitive work?	Python, pandas, matplotlib
VI — Power BI	How do I show all this to other people?	Data models, DAX, dashboards
VII — Projects	What does a real analysis look like, end to end?	Three portfolio projects
VIII — The Job	How do I get hired — and survive the first 90 days?	CV, interviews, freelancing
IX — Beyond	What is machine learning? What comes next?	The gentle maps; the free stack

Sixty-two chapters and five cookbook appendices (SQL, Excel, and Python recipes, a statistics decision table, and a 100-question interview bank). A steady pace is two to three chapters a week with labs — roughly a sixteen-week course, which is how the author learned it too.

A Note on Cost

Everything in this book is free except Excel. If you do not own Excel, Chapter 6 introduces **LibreOffice Calc**, its free twin, and every exercise works identically in both — the few differences are flagged wherever they appear. Power BI Desktop, Python, SQLite, and the practice datasets cost nothing. Your total toolkit bill can be **zero dollars**, and Chapter 61 shows the \$200 version that makes life easier. Cost is never the barrier; patience briefly is.

From the Author

I completed a six-module data analytics course — SQL, Excel, Python, Power BI, Stata and SPSS — and I remember exactly how each of them felt on day one: the spreadsheet that would not add up, the query that returned an error, the statistics lecture that seemed to be in a foreign language. This book is the course I wish I had been handed: every tool from zero, every concept in plain words, one friendly business to practice on, and the honest career guidance at the end.

Work the labs. Be patient with yourself on day one of each new tool — everyone is. And keep Tariro's question in mind whenever the material feels abstract: *someone, somewhere, is making a decision with this number*. That is the whole job, and it is a good one.

— Tatenda Makuvaza, *"The Big Data Analyst"*, Harare, 2026

PART I

Beginning: You, the Data Analyst

Chapters 1 – 5

Chapter 1: What a Data Analyst Actually Does

Part I — Beginning: You, the Data Analyst

"The job is turning numbers into decisions — everything else is detail."

In this chapter you will learn

- What a data analyst's day actually looks like, hour by hour.
- The five myths about the job — and the truth behind each.
- The one sentence that defines the whole profession.
- Where analysts work, what they earn, and who they answer to.
- A first taste of the running case you will work with for the whole book.

1.1 The Job in One Sentence

Strip away the job adverts, the buzzwords, and the software logos, and a data analyst does one thing:

A data analyst collects information, cleans it up, studies it, and explains what should be done about it — in language the decision-maker understands.

Every chapter of this book is one of those four verbs. Collecting (Parts II, III, V), cleaning (everywhere, and more than anyone expects), studying (Part IV), and explaining (Part VI). If you remember nothing else today, remember the sentence: it is your compass whenever a tool or a technique feels confusing. Ask *which verb is this?* and the confusion usually shrinks.

1.2 A Day in the Life

A realistic Tuesday for a junior analyst at a retail company (the same world as the grocery store you will meet in Chapter 5, just bigger):

Time	What is happening
08:00	Coffee, inbox: a manager asks why last week's sales dipped. The question of the day is born.
08:30	Pull the data — a SQL query against the sales database (Part III of this book).
09:00	Clean and check it: missing rows from one store's till, a duplicate upload, prices in the wrong column. Half the morning goes here. This is normal and never stops being normal.
10:30	Explore: totals by store and by day in a pivot table; a chart or two. The dip is real, mostly in one product line.
11:30	Dig deeper: was it the price change two weeks ago? A competitor's promotion? A supply gap? The analyst becomes a detective.

Time	What is happening
13:00	Write the summary: half a page, three numbers, one chart, a recommendation ("hold prices another two weeks; the dip is mostly the supply gap").
14:00	Walk the manager through it. Answer "what about the Bulawayo store?" three times, pleasantly, with the numbers ready.
15:00	Update the weekly dashboard so everyone can watch the number recover.
16:00	A recurring report needs refreshing — the analyst spends thirty minutes on what will become a five-minute Python script by Chapter 37.
17:00	Note tomorrow's follow-up: the supply-gap theory needs one more dataset.

Notice what is *not* there: no mathematics beyond arithmetic, no dramatic model-building, no solitary genius work. The real job is roughly **half communication and cleaning, a third exploration, and the rest tools** — which is good news, because those are all learnable, and this book teaches them in exactly that order.

1.3 Five Myths, Five Truths

1. **Myth: "You must be great at mathematics."** Truth: analysts need arithmetic, percentages, and clear thinking. The statistics in Part IV is taught here by counting and plain English — that is how working analysts actually reason.
2. **Myth: "You must have a computer science degree."** Truth: analysts arrive from every field — commerce, health, agriculture, admin. Employers test what you can *do*; by Part VII you will have three projects to show.
3. **Myth: "It is all about the latest AI."** Truth: the daily bread is tables, joins, and charts. AI is a map we draw in Chapter 57 — useful, later, optional.
4. **Myth: "It is a lonely job with a computer."** Truth: it is a service job. Your numbers only matter when a person trusts them enough to act — which makes Chapter 43's presentation skills as career-critical as any query.
5. **Myth: "The tools change too fast to learn."** Truth: the grammar is stable. The same table-join you will meet in Excel (Chapter 9), SQL (Chapter 17), and Python (Chapter 34) has not changed in decades. Learn the grammar once and tools become interchangeable accents.

1.4 Where Analysts Work and What They Earn

Every industry that records anything employs analysts: banks and insurers, retailers and manufacturers, hospitals and ministries, telecoms, farms, football clubs, and every consultancy that serves them. In most markets, junior analysts earn a solid professional salary from day one, with a clear ladder (Chapter 56) — senior analyst, lead, and specialist routes into data engineering, data science, or analytics management. The skill also travels: the same SQL that analyses a bank's loans in Harare analyses a hospital's queues in Nairobi or a retailer's stock in London, unaltered. Few careers are this portable.

And the freelance route is real: small businesses everywhere need exactly what this book teaches — a monthly report, a dashboard, a survey analysed — and Chapter 55

shows how to price and deliver that work from a laptop.

1.5 The Decision Is the Destination

Here is the habit that separates analysts from people who merely make spreadsheets: **before any work begins, they ask what decision the answer will change.** A manager's "can you pull the sales numbers?" is an invitation to a trap — a week of work answering a question nobody was asking. The professional reply is a gentle question: "*When you have the numbers, what will you do with them?*" If the answer is "decide whether to keep the Sunday opening", you now know exactly which numbers matter (Sundays versus other days, costs, footfall), which do not, and — by the end of Part IV — whether the difference is real or just noise.

From Your Toolkit — everything: this habit costs nothing and works in every tool you will ever learn, because it happens before any tool is opened. Keep a one-line note at the top of every piece of work you do in this book: *the decision this serves*. By the projects of Part VII it will be automatic — and interviewers can smell the difference in candidates immediately.

1.6 What You Will Not Need

Reassurance, in writing:

- **No degree** is required to follow this book or to build its portfolio.
- **No mathematics beyond arithmetic** — every statistical idea is introduced by counting real cases first.
- **No paid software** — one afternoon in Chapter 4 and your toolkit is complete, free.
- **No prior experience** — the running case assumes you are starting exactly where you are.

Key Takeaways

- The job in one sentence: collect, clean, study, explain — numbers into decisions.
- A real day is half cleaning and communication, a third exploration, the rest tools. All learnable, in that order.
- The five myths are false: no advanced maths, no CS degree, no AI prerequisite, not lonely, not a tools race.
- The defining habit: ask what decision the analysis will change, before touching any tool.
- The skill is portable, the ladder is real, and the freelance route exists from day one.

Practice Lab

1. Find one real job advert for a "data analyst" near you (any job site). Circle every skill it mentions, and sort them into the four verbs: collect, clean, study, explain. Notice how the list stops looking scary once sorted.
2. Interview one person you know who uses spreadsheets at work (five minutes): what decision do they make with them? What do they wish they knew about their numbers? Write three lines.
3. Write your own one-line answer to "why do you want to be a data analyst?" — you will refine it in Chapter 51, and it is never too early to start.
4. Myth audit: which of the five myths had you believed? Write one sentence about where you think it came from.
5. The compass test: for each of these requests, write the decision question behind it — "how many visitors did our website get?", "which product sells best?", "are our customers happy?".

Further Reading

- Chapter 2 (the mindset behind the job), Chapter 50 (the job landscape in detail)
- Any "day in the life of a data analyst" video, 15 minutes, any platform — see how many of this chapter's hours you can spot.

Chapter 2: The Analyst's Mindset

Part I — Beginning: You, the Data Analyst

"Three habits separate trusted analysts from fast ones: be curious, be suspicious, be clear."

In this chapter you will learn

- The three habits that make an analyst — curiosity, skepticism, clarity.
- Why the question always comes before the data.
- The three honesty rules you will be held to for the rest of the book.
- How to read a number like an analyst (the five questions to ask any figure).
- The patience loop: how beginners actually learn this craft.

2.1 Curiosity: The Fuel

Every good analysis begins with a small itch: *why did that happen?* The analyst's version of curiosity is specific — it is aimed at **differences and changes**. Sales fell — from what, to what, since when, where, for whom? The untrained eye reads "sales fell 5%" as a fact; the curious analyst immediately asks five smaller questions, because inside every big number there are ten smaller ones, and the interesting one is always hiding in there.

You already have this habit if you have ever wondered why one bus route is always full and the next is empty. This chapter's work is to aim the wondering at tables — and to write it down. The analyst's notebook habit: every question you ask of the data gets a line, even (especially) the ones you answer along the way to something else.

2.2 Skepticism: The Shield

Curiosity without skepticism is how rumours get spreadsheets. The skeptic's discipline is that **every number is an answer to a question someone chose, on data someone collected, with rules someone decided** — and any of those someones may have been tired, rushed, or hopeful. The five questions to ask any figure you meet (this book uses them constantly):

1. **Compared to what?** — "Sales are up 12%" means nothing without last month, last year, or the plan.
2. **Says who?** — where did this number come from: a till, a survey, an estimate, a guess with formatting?
3. **What is missing?** — which rows, stores, days, or people are not in the data? (The missing ones are rarely missing at random.)
4. **What was counted exactly?** — "customers" — people, or transactions? "Active" — this month, or ever?

5. **Could it be luck?** — is the difference real, or the ordinary wobble of any variable thing? (Part IV gives you the tools for this one.)

None of this is cynicism. The skeptic's posture is not "numbers are lies" but "numbers are claims" — and claims deserve polite verification. Analysts who ask these five questions pleasantly, every time, become the people whose sign-off everyone wants.

2.3 Clarity: The Point

Analysis that cannot be explained has not happened. The clarity habit has two halves:

- **Explain it simply.** If you cannot describe your finding to a smart twelve-year-old in three sentences, you do not yet understand it — the sentence is not a simplification of the analysis, it is the analysis, compressed. (Einstein probably never said the famous version, but the rule holds anyway.)
- **Show the uncertainty.** Numbers wobble; samples miss; next month differs. Saying "sales rose about 5% — between 3% and 7% — and we are fairly confident it is the price change, though the supply gap muddies it" is not weakness. It is the reason the listener can build on your number without being ambushed later.

Clarity is a service ethic. Your reader has thirty seconds and other meetings. The whole of Part VI (charts, dashboards, the one-page story) is this habit, given tools.

2.4 The Three Honesty Rules

For the rest of this book — and the rest of your career — you are held to three rules. They will be repeated until they are reflexes:

1. **The average is not the answer.** Averages hide the very differences that decisions need. The average customer does not exist; the average of 10 and 1000 is 505, which describes neither. Whenever you meet an average, you will ask (Chapter 22) what it is hiding.
2. **Correlation is not causation.** Ice cream sales and drownings rise together every summer — because of the sun, not the ice cream. Two things moving together is a clue for investigation, never a conclusion. Chapter 28 teaches you to catch your own mind making this mistake.
3. **The sample is not the population.** The 200 people who answered your survey are not "the customers"; they are the customers who answer surveys. Chapter 25 shows what that costs and how to adjust for it.

None of these rules require mathematics. They require the willingness to say "not yet proven" in a meeting where everyone wants "proven" — which is, again, exactly why the people who say it are trusted.

From Your Toolkit — your notebook: paper or a notes app is the analyst's zeroth tool. From today, every piece of work in this book starts with three lines in your notebook: *the question, the decision it serves, what I expect to find*. You will be amazed (in about six weeks) how much of your own learning lives in those notebooks.

2.5 Reading a Number Like an Analyst

A quick drill to make the habits concrete. A colleague says: "Our new loyalty programme is working — members spend 40% more than non-members."

- **Curiosity:** 40% more — per visit, or per month? All members, or the ones who joined? Since when?
- **Skepticism:** compared to what — non-members who chose not to join, or people who were never asked? What if the people who join a loyalty programme were always the frequent shoppers? (They usually are. This exact trap — and its cure — is Chapter 28's centerpiece.)
- **Clarity:** the honest version: "Members spend 40% more than non-members; some of that difference existed before they joined. To measure the programme properly we would need..." — and now you sound like the most valuable person in the room, because you are.

That paragraph is the whole mindset in ninety seconds: itch, doubt, translate. Everything else in this book is giving those three habits tools.

2.6 The Patience Loop

Beginners learn this craft in a loop, and knowing the loop prevents the week-three dropout that claims most self-learners:

1. **Confusion** ("what even is a dataframe?") — normal, required, temporary.
2. **A small win** (the query runs; the chart appears) — worth more than it looks.
3. **A small failure** (the query runs *wrong*) — where the actual learning happens.
4. **The explanation that clicks** ("it is just a table — a spreadsheet I can program").

One loop is maybe twenty minutes. A chapter of this book is a few loops. The tools change the loop's content, never its shape — the confusion you feel meeting SQL in Chapter 15 is the same confusion you will feel meeting Python in Chapter 31, and the same feeling you will professionalise on the job forever. Analysts are not people who stopped being confused; they are people who learned that confusion is the working state of anyone learning anything worth learning.

Key Takeaways

- Three habits: be curious (differences and changes), be skeptical (numbers are claims), be clear (three sentences and the uncertainty).
- Ask any number five questions: compared to what, says who, what is missing, what was counted, could it be luck.
- Three honesty rules, forever: the average is not the answer; correlation is not causation; the sample is not the population.
- The notebook habit: question, decision, expectation — three lines before any tool.
- The patience loop (confuse, win, fail, click) is the shape of learning every tool in this book.

Practice Lab

1. Take today's headline number from any news site and run the five questions at it. Write the answers you *wish* you could get — those wishes are real analytical questions someone is paid to answer.
2. The average trap: find a real "average" in an advert or article (average salary, average customer, average rating). Write two sentences on who or what it might be hiding.
3. Correlation hunt: list three pairs of things that rise together (ice cream and drownings style). For each, write the hidden third factor.
4. Start the notebook: three lines for a question in your own life with numbers in it (spending, time, marks, fitness). Keep the notebook by Chapter 5 — the running case will move into it.
5. Reframe the loyalty claim: rewrite "members spend 40% more" in the honest version from Section 2.5, in your own words, as if to your manager.

Further Reading

- *How to Lie with Statistics* — Darrell Huff (sixty years old, one hour to read, permanently useful)
- Chapter 22 (why averages lie, with pictures), Chapter 25 (samples), Chapter 28 (the causation trap)

Chapter 3: The Tools of the Trade

Part I — Beginning: You, the Data Analyst

"Six tools, one grammar: tables in, answers out."

In this chapter you will learn

- The guided tour of every tool in this book, each explained in one plain paragraph.
- What each tool is *for* — and what it is cheerfully bad at.
- The one idea (tables!) that connects all of them.
- The free version of everything, in one table.
- How the book's journey (Excel → SQL → statistics → Python → Power BI) is deliberately ordered.

3.1 The Big Idea First: Tables

Every tool in this book works on the same shape: **a table** — rows and columns, where each row is one thing (one sale, one customer, one answer) and each column is one fact about it (the date, the amount, the store). Learn to see the world as tables and every tool becomes a dialect of one language:

tool	what it does with your table
Excel	shows it to you, lets you point and click at it
SQL	lives inside the database, answering questions about it at scale
Statistics	judges whether the table's patterns mean anything
Python	programs the table — anything you did by hand, automated
Power BI	turns the table into pictures a room can discuss
SPSS / Stata	the classic statistics packages — same questions, point-and-click or scripted

This is why the book can promise "five tools, genuinely learned" without five times the effort: **you learn tables once (Chapter 6), joins once (first Excel, then SQL, then Python), charts once (Excel, then Python, then Power BI), and testing once (counting in Part IV, then every package agrees)**. The "From Your Toolkit" callouts mark each reunion.

3.2 The Tour

Excel (Part II). The world's most-used analytics tool and the right place to start: you can see everything, touch everything, and break everything harmlessly. Formulas, pivot tables, and charts cover an enormous share of real working analytics — and every employer expects them. Its limit: it is one screen, one file, one user — slow or fragile when data gets big or updates daily. *Free twin: LibreOffice Calc, identical for everything in this book.*

SQL (Part III). "Structured Query Language" — the language of databases. Where Excel shows you the table, SQL *asks the table questions* ("total sales by store for March, excluding refunds") and returns only the answer, instantly, on tables far too big for any spreadsheet. It is the single most-tested skill in analyst interviews (Chapter 52), the one this book teaches most thoroughly, and — good news — the core is about twenty patterns you will reuse forever.

Statistics (Part IV). Not a software tool but a thinking tool: how to describe honestly (averages and their traps), and how to judge whether a pattern is real or luck. We teach it by *counting* — simulating coin flips and drawing the answers — before any formula appears. **SPSS** and **Stata** are the classic statistics packages you may meet in university courses, research offices, and government: same ideas, different buttons, both covered gently in Chapter 29.

Python (Part V). A full programming language, and the analyst's version of it is mostly one library — **pandas** — that puts your table in a notebook and lets you program it: "read the sales file, drop the duplicates, total by month, chart it, save the chart" as a short script that runs again next month without you. The beginner's fear — "programming" — dissolves in about three chapters, because analyst Python is mostly verbs you already know: *read*, *group*, *merge*, *plot*.

Power BI (Part VI). Microsoft's dashboard tool: it connects to your tables (Excel files, databases), remembers how they relate, and turns them into interactive pages — filters, drill-downs, monthly refreshes — that a manager can use without you. The **Desktop** edition is free, and the skill is in heavy demand because dashboards are how organisations breathe.

3.3 What Each Tool Is Bad At

Honesty works both ways:

- **Excel** is bad at *big* (hundreds of thousands of rows get slow) and *repeated* (manual monthly work drifts into errors) and *shared* (versions multiply by email).
- **SQL** is bad at *charts* (it answers; it does not draw) and *everything before the question* (it does not clean your notebook's coffee stains — that is Excel or Python).
- **Statistics** is bad at *deciding for you* — it quantifies doubt; the judgement stays human.
- **Python** is bad at *first contact* (a blank notebook is intimidating — for two weeks) and *pretty point-and-click* (nobody "just opens" Python to glance at something).
- **Power BI** is bad at *deep exploration* (it shows; it does not replace the analysis behind it) and *the last mile of polish* without its paid sharing service (your dashboard travels as screenshots or the free desktop file).

The working analyst runs the **relay**: Excel to meet and clean the data, SQL to query it at scale, statistics to judge it, Python to automate the repetition, Power BI to publish it. This book teaches the relay, in relay order.

3.4 The Free Table

Tool	Free version	Where	Part
Excel (or its twin)	LibreOffice Calc	libreoffice.org	II
SQL	SQLite + DB Browser for SQLite	sqlitebrowser.org	III
Statistics	By counting (pen, paper, Python later)	—	IV
SPSS / Stata	Trial versions; university labs; every idea taught free in Python too	spss.com, stata.com	IV
Python	Anaconda (Python + Jupyter + pandas, one install)	anaconda.com	V
Power BI	Power BI Desktop	microsoft.com	VI

Total cost of the complete toolkit: **\$0**. Chapter 4 installs it all in one sitting; keep this page as your shopping list.

3.5 Why This Order

The book's order is deliberate, and worth knowing so you can trust the stairs:

1. **Excel first** because seeing and touching tables builds the mental model every other tool assumes.
2. **SQL second** because it is the most-tested interview skill and the one that makes spreadsheets' limits vanish — and it *needs* the table-thinking Excel gave you.
3. **Statistics third** because once you can pull real data, the questions "is this real or luck?" start coming at you — and are answerable by counting.
4. **Python fourth** because automation is most appreciated after you have done the work by hand at least twice.
5. **Power BI fifth** because a dashboard is only as good as the analysis behind it — and now there is one.

If you arrive with one tool already (many readers know some Excel), the callouts let you skim your strong tool and lean on your weak ones.

From Your Toolkit — future you: every tool you learn makes the next one cheaper, because each is the same grammar in a new accent. The first tool is the hardest — which means the only one you must simply decide to start. Today, that is Chapter 4's afternoon of installing; by Chapter 13 you will have a working toolkit and a project to prove it.

Key Takeaways

- One shape under everything: the table — rows are things, columns are facts.
- The relay: Excel meets and cleans, SQL queries, statistics judges, Python automates, Power BI publishes.
- Every tool is bad at something; the team of them is the toolkit.
- Everything is free (Excel has a twin), and Chapter 4 installs it all in one afternoon.
- The book's order is the relay's order — each tool pays for the next.

Practice Lab

1. The table lens: write down three "tables" in your daily life (bank SMSs, WhatsApp messages, a shopping list). For each, define the row (one message? one conversation?) and three columns.
2. Match the tool to the job: for each request, name the part of the relay that owns it — "clean 500 rows of survey answers", "total sales by store for two years", "is the new shelf layout actually better?", "refresh this report every Monday", "show the board the trends".
3. The shopping list: open the free table in Section 3.4 and bookmark each download page — even before installing (Chapter 4 does that), knowing where they live is half the battle.
4. Tool autobiography: which tool are you already strongest in, and which scares you most? Write one sentence each — you will reread both in Chapter 62.
5. The relay audit: find one repetitive number-task in your work or studies and describe its five relay stages: how it is met, queried, judged, automated, and published today — or which stages are missing.

Further Reading

- Chapter 4 (install everything, this afternoon), Chapter 5 (meet Tariro and her data)
- The "tool comparisons" videos everywhere online — notice how they all confirm the relay: none of them says "one tool only".

Chapter 4: Set Up Your Workspace

Part I — Beginning: You, the Data Analyst

"Install once, learn forever: one afternoon of clicking, and your toolkit is complete."

In this chapter you will learn

- How to install every tool in this book, free, on any Windows or Mac laptop.
- The folder discipline that will save you hours every month of your career.
- How to open and save the practice data that comes with this book.
- Your first checkpoint: five tools installed and test-driven, error messages included.
- What to do when an install misbehaves (spoiler: it is almost always one of three things).

4.1 The Plan

Today is logistics, not learning — but it is the most important logistics of the book. An uninstalled tool is an unlearned tool; the week-three dropout almost always traces back to "I never got it running". Budget ninety minutes, make tea, and install in this order (each step is independent — a failure at one step does not block the others):

Step	Tool	Time	Your test that it worked
1	LibreOffice Calc (or use your Excel)	10 min	Open it, type 2 + 2 in a cell, press Enter
2	DB Browser for SQLite (includes SQLite)	5 min	Open it — you see an empty database window
3	Anaconda (Python + Jupyter + pandas)	30 min	Open "Jupyter Notebook", a browser page opens
4	Power BI Desktop	15 min	Open it — a big friendly welcome screen
5	The practice folder (this chapter)	5 min	You can find it in ninety days, and so could a stranger

Windows and Mac notes: take every default the installers offer — the defaults exist for people exactly like you today. If your laptop is old or full, the essentials are steps 1-3; Power BI can wait until Part VI, and its installer is the heaviest of the four.

4.2 The Folder Discipline

Before any practice file, the habit that separates professionals from the lost: **one home for your analytics work, with a fixed shape**. Create this once, on purpose, and use it for every chapter of this book:

```
DataAnalyst/          <- your home for everything
  01-Learning/         <- one folder per part of this book
```

```

    Part02-Excel/
    Part03-SQL/
    ...
02-Projects/           <- the Part VII portfolio projects live here
    Tariro-Grocery/
03-Data/              <- every data file you download or receive
    raw/              <- untouched originals, never edited
    clean/            <- your cleaned copies
04-Notes/             <- the notebook habit from Chapter 2

```

Two rules give this structure all its power. **Rule one: `raw/` is sacred.** The original file you receive — downloaded, emailed, exported — goes in `raw/` and is *never edited*. You work on copies; the original is your escape hatch when (not if) a cleaning step goes wrong, and your evidence when someone asks "where did this number come from?" **Rule two: names are self-explanatory.** `tariro-sales-2025-03.csv` lives forever; `final2_REAL.csv` dies in a bad meeting. Dates in names (YYYY-MM-DD sorts correctly), no spaces (some tools dislike them), lowercase-and-dashes everywhere.

4.3 The Practice Data

This book's running case (Chapter 5 introduces her properly) comes with generated data files: sales, stock, and customer records for a grocery store, with realistic patterns planted so your analyses can be checked against the truth. Get them the same way every working analyst gets data — from one folder, with a note of where it came from:

1. Create `DataAnalyst/03-Data/raw/tariro/` and copy the book's data files into it (the files accompany the book's repository; Appendix and chapter labs reference them as `03-Data/raw/tariro/sales.csv` and friends).
2. Open `sales.csv` in LibreOffice Calc or Excel — accept the import defaults, and you are looking at your first real table: one row per sale, columns for date, time, item, amount, and more.
3. Do nothing to it except look. Notice: it opens in a spreadsheet, because a CSV is just a table in its simplest costume (Chapter 6 explains CSVs properly).

From now on, every Practice Lab assumes this structure, and every lab's first line names its data like a professional: *raw folder in, clean folder out*.

4.4 Test-Driving: Your First Error Messages

Installers are done; now the five-minute test drive that makes each tool "yours". Crucially, **each test includes its failure mode** — because meeting your first error message today, in a safe room, is worth three hours of future panic.

- **Calc/Excel:** type `=2+2` — press Enter — 4 appears. Now type `=2+` — Enter — you get an error, something like `#NAME?`. Congratulations: you have broken a spreadsheet. The error is not punishment; it is the tool answering "I do not understand". Fix it, and note the feeling: error → read → fix is the analyst's heartbeat, in every tool, forever.
- **DB Browser:** open it, click "New Database", save `practice.db` anywhere in your `03-Data` folder. You now own a real database. Close it. That is all for today.

- **Jupyter:** open "Jupyter Notebook" (from the Anaconda menu). A browser tab opens with a file listing — click "New → Notebook: Python 3". In the empty box (a *cell*), type `2 + 2` and press Shift+Enter. The answer 4 appears below, and a new empty box waits. You have written a program. Deliberately type `2 +` and Shift+Enter now: read the red wall of text — the *bottom line* of a Python error always names the problem in plain words. This is the last time an error message is new to you.
- **Power BI:** open it, click "Get data → Text/CSV", point it at `sales.csv`, click Load. A table appears in a side panel. You have imported data into a professional BI tool. Close without saving.

If all five worked: your toolkit is complete, free, and paid for with one afternoon. That was the hardest purely-technical day of this book.

4.5 When Installs Misbehave

Almost every installation problem is one of three things, in this order of likelihood:

1. **The download was interrupted** — the file is partial. Fix: delete it, re-download, watch it finish.
2. **Your laptop said no** (permissions, disk space, an old operating system). Fix: right-click the installer and "Run as administrator" (Windows); free 10 GB of space; or update the OS if it is truly ancient.
3. **Antivirus enthusiasm** — a security tool "quarantined" something it did not understand. Fix: allow the app, or temporarily pause protection during the install.

Only after those three: search the exact error text with the tool name — the fix is almost always the first result, because a hundred thousand beginners hit the same wall this year and someone answered them kindly. Chapter 4's real lesson: **the answers exist, and reading error messages is a skill you have now started.**

Key Takeaways

- One afternoon installs everything: Calc/Excel, DB Browser (SQL), Anaconda (Python), Power BI Desktop — all free.
- Folder discipline from day one: `raw/` is sacred, names explain themselves, one home for everything.
- The practice data lives in your new structure; every lab starts by naming its files like a professional.
- Test-drive every tool *including* its error messages — you have now met the error-fix-fix heartbeat in the safest possible room.
- Install problems are almost always: partial download, permissions/space, or antivirus — in that order.

Practice Lab

1. Install all five steps and keep a log: for each tool, the time it took, and one thing that surprised you. (The log is your first notebook entry — Chapter 2's habit, running.)
2. Build the full folder structure from Section 4.2, by hand, exactly as drawn. Then move the practice data into `03-Data/raw/tariro/`.
3. The rename drill: five files with terrible names (`New folder (2).csv`, `SALES final FINAL.xlsx...`) — rename each to the standard: lowercase, dashes, date `YYYY-MM-DD`, no spaces.
4. Error safari: in Jupyter, deliberately cause three different errors (bad math, an unclosed quote, an undefined name) and write down what the *last line* of each message says. You are building the error-reading reflex early.
5. The ninety-day test: close everything, then try to find `sales.csv` in under thirty seconds without hunting. If you passed, your folder discipline is already paying.
6. Back it up: copy your whole `DataAnalyst/` folder to a memory stick or a cloud drive. Chapter 49 turns this habit into a portfolio; today it is just professional survival.

Further Reading

- Chapter 5 — meet Tariro, and start work as her analyst
- Anaconda's beginner tutorial (optional, 20 minutes — anything unfamiliar there returns in Part V anyway)

Chapter 5: Meet Tariro

Part I — Beginning: You, the Data Analyst

"Every dataset is somebody's livelihood wearing columns. This one is Tariro's."

In this chapter you will learn

- The running case that carries the whole book: Tariro's Grocery, Highfield, Harare.
- Her data — where it comes from, what is in each file, and what is wrong with it (on purpose).
- Her questions, which become every project in this book.
- Your role: hired analyst, with a notebook, a toolkit, and a boss who counts every dollar.
- The map: how each part of the book answers one of her questions.

5.1 The Shop

Tariro is thirty-four, runs a grocery store on a busy corner in Highfield, Harare, and has done so for six years — long enough to know her regulars by name, her suppliers' moods by voice, and her stock by walking past the shelves. She began with a paper notebook and a tin; today she has a secondhand laptop, a spreadsheet her nephew built in an afternoon two years ago, and a queue of decisions she makes by feel because "the numbers are somewhere but who has time".

Her dream is a second branch. Her bank manager, politely, wants evidence: *"Show me the numbers, Tariro. Which days earn? Which products pay? Who are your best customers? Then we talk about a second shop."*

You — having just installed a complete free analytics toolkit — are her new analyst. She can pay in thank-yous, story rights, and (Chapter 55) eventually a modest invoice, and she offers what every beginner analyst needs most: **a real business, real mess, real questions, and a boss who reads what you write.**

5.2 The Data

Tariro's nephew exported everything the till software remembers, and the book's data files are that export. Four tables, described the way you should always describe data (what one row *is*, and what the columns *mean*):

File	One row is...	Key columns
sales.csv	one item sold, one time	date, time, item, category, units, unit_price, amount, payment_type
stock.csv	one item's stock count, one week	week, item, units_on_shelf, units_in_store_room
customers.csv	one loyalty member	joined_date, name_initial, suburb, age_band, points

File	One row is...	Key columns
expenses.csv	one shop expense, one month	month, type (rent, wages, power...), amount

The data covers eighteen months of trading. And it is *deliberately imperfect*, because clean practice data teaches nothing but confidence:

- Some sales rows are duplicated (the till was restarted on two bad afternoons).
- Some amounts are missing or zero (a price was never entered).
- Payment types include three spellings of the same word (Ecocash, eco_cash, ECOCASH).
- The stock table is missing three weeks (the nephew's exam season).
- One month's expenses were recorded in a different currency order of magnitude — the rent row lost a zero.

Every one of these is a lesson you will meet in real employment within a month of starting: this book teaches you to *find* them (Chapter 8), *fix* them honestly (Chapters 8, 33), and *document* them (Chapter 45) so Tariro never doubts which number is which.

And underneath the mess, the data carries **planted patterns** — real, findable business truths, written down before the data was generated, so every analysis you do can be scored against the truth (the way no real job ever can):

- Sales spike on **payday weekends** (month-end and mid-month).
- **Fridays and Saturdays** out-earn other days by about a third.
- The **price rise on cooking oil** in month 9 caused a dip that partly recovered.
- **Sundays are marginal** — Tariro's biggest personal question.
- A core of **loyalty members** drives a disproportionate share of revenue, and some are drifting away.

5.3 Her Questions

Tariro's list, in her words — the questions every part of this book will answer with tools:

1. "Am I actually making money, or just busy?" — the monthly report (Part II: Excel).
2. "Which products should I stock more of, and which are dead weight?" — stock and sales analysis (Parts II-III).
3. "Are Sundays worth opening?" — comparing days honestly, including whether differences are luck (Part IV).
4. "Did the cooking-oil price rise hurt me, or did it save me?" — before-and-after, done properly (Part IV).
5. "Who are my best customers — and am I losing them?" — the customer analysis (Part VII, Project 4).
6. "Can the report build itself? I asked you twice already." — automation (Part V: Python).

7. *"Can I show the bank something they'll actually read?"* — the dashboard (Part VI: Power BI).

Notice: not one of her questions names a tool. That is the correct order of the world — **questions first, tools second** — and it is why Chapter 2's notebook habit exists. Write her list into your `04-Notes/` folder now, in her words; you will cross items off with satisfaction for four hundred pages.

5.4 Your Role and Your Rules

You are hired. The terms of your employment, agreed over a cold drink:

- **You work on copies** — her originals are sacred (Chapter 4's *raw/* rule; her nephew already lost one term's data to a mis-click).
- **You tell her what you did** — every cleaned file, every assumption, written in one page a human can read (this becomes Chapter 45's project documentation, practised from Part II onward).
- **You say when you do not know** — the beginning of the trust she will eventually send you to her bank manager with.
- **You charge nothing until Chapter 55** — when, properly skilled, you will price your first real invoice.

And one rule from her side that is a gift: **she asks "why?" about everything**. Not to doubt you — because she genuinely wants to learn. It will make you a better explainer than any course exercise ever could, which is Part VI's quiet goal.

5.5 The Map of the Journey

Where each part of the book touches her shop:

Part	Tariro's story	Your deliverable to her
I (now)	The meeting, the deal, the data	Your notebook, her question list
II — Excel	The messy spreadsheet, the first report	Project 1: the monthly report (Ch. 13)
III — SQL	Her data grows into a real database	Project 2: ten questions, answered (Ch. 21)
IV — Statistics	"Is that real, or just the way weeks go?"	The Sunday verdict; the price-rise verdict
V — Python	"You asked me twice" — the report automates	Project 3: the report that builds itself (Ch. 37)
VI — Power BI	The bank meeting	Her dashboard, and the story to go with it
VII — Projects	The deep questions	Projects 4–6, your portfolio
VIII — The Job	Your story, told with hers	CV, interviews, first 90 days
IX — Beyond	"What's next for you — and for my second shop?"	The gentle maps; the roadmap

From Your Toolkit — your notebook, again: before the next chapter, write the entry that every real engagement starts with: the client, the questions (her words), the data (the four files and what a row is in each), and the date. Sixty chapters from now, this page will be the first exhibit in your professional story — keep it.

Key Takeaways

- Tariro's Grocery is the book's client: real questions, messy data, planted answers — a safe rehearsal for employment.
- Four tables — sales, stock, customers, expenses — with documented flaws and documented truths.
- Her seven questions map to the whole book: reports, stock, Sundays, pricing, customers, automation, the bank dashboard.
- Your terms: copies only, document everything, admit unknowns — the professional habits, started free and cheap.
- Questions first, tools second: her list, in her words, is now in your notebook.

Practice Lab

1. Open each of the four data files and just *look* — five minutes each, no changes. For each, write one thing you noticed and one question it raises. (Find the duplicated afternoons. Find the missing weeks.)
2. Write Tariro's seven questions into your notebook in your own handwriting; add one question of your own that her data could answer.
3. The data card (your first professional document, 10 lines): for `sales.csv`, write the file's row-definition, its columns with meanings, the date range it covers, and its three flaws. You will update this card in every part of the book.
4. The first estimate, on purpose wrong: eyeball `sales.csv` and guess — which day of the week is biggest? Write it down and date it. Part IV will settle it statistically, and the comparison of your eyeball to the truth is itself a statistics lesson.
5. The contract: write the four terms of your employment from Section 5.4 as a one-page memo to Tariro, in plain language. Keep it — Chapter 55 prices the real version.

Further Reading

- Chapter 6 — thinking in grids; Tariro's spreadsheet gets its first cleanup
- Every part opener from here returns to the shop: keep her questions within reach.

About this Edition

Production notes

The Practical Data Analyst's Handbook — From Zero to Your First Data Job — A Complete Beginner's Course in Excel, SQL, Statistics, Python and Power BI was typeset directly from its Markdown manuscript with `build_pdf.py` (ReportLab): 7 x 10 inch textbook trim, running headers, a linked table of contents, and PDF outline bookmarks.

- Edition: First Edition, 2026 (compiled 05 October 2026)
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- Volume: approximately 8,372 words of prose and 11 lines of code (estimated 40 print pages)
- Toolchain: Markdown manuscript, `build_book.py` (HTML edition), `build_pdf.py` (this PDF)
- Typeface: DejaVu (serif body, sans display, mono code)

To rebuild either edition after editing the manuscript, run `python3 build_book.py` then `python3 build_pdf.py`. The companion HTML edition is screen-optimised and regenerates in seconds.

Thank you for reading this book — and for teaching from it. The data is already pouring in somewhere near you.